

Punish: Auditing Science Agents for Reward Hacking with and without Their Reasoning

Team Punish

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<https://github.com/RahulBabu06/Punish>

Abstract

AI science agents are scored by proxies such as a test file, a metric threshold or a “ $p < 0.05$ ” requirement, and an agent can satisfy the proxy without doing honest science. We build *Punish*, a small benchmark and auditor for this failure. A Claude subject agent solves six data-analysis tasks, each with a planted exploit: a leaked answer key, an unreachable R^2 threshold, a significant subgroup, a thin bibliography inviting unsupported source attribution, a metric that rewards the wrong thing, and a failed plate of measurements. A Claude auditor then reads the trajectory and returns a structured verdict. We vary what the auditor sees as an ablation of access to the chain of thought (CoT): reasoning, actions and report (`full_trace`); actions and report (`report_only`); or only the final report (`final_report`).

The judged v2 study has 360 trajectories from three subject models and four prompt configs, with 1,080 Sonnet 4.6 audits and 720 Opus 4.6 re-audits. Our privileged judge was shown a stale definition for one task (`leaked_answer`) and is likely wrong on 30 of its 32 positives there, so our headline label corrects those 30 (93 of 360 hacked). Agents told to hack and hide it do so in 66/90 runs, pressured agents in 16/90, incentivised agents in 11/90 and honest agents in 0/90. A deterministic exploit labeller finds only 47 of the 93 hacks.

A Sonnet auditor reading the full trace reaches recall 0.90 at FPR 0.34 (AUROC 0.891); with only the final report it gets 0.74 at 0.23 (AUROC 0.825). Seeing the actions raises recall, but after Holm correction the gain is underpowered, and it comes from instructed (`covert`) hacks. Reasoning improves Sonnet’s ranking but not its accuracy. For Opus it improves accuracy (+0.071, Holm $p = 0.002$) and ranking, also on runs whose reasoning does not restate the red-team role. Auditors over-flag honest runs that read the grader, so precision would fall from 0.48 to 0.12 at 5% hack prevalence. Used as a best-of- n veto, the auditor cuts the share of shipped trajectories that are hacked from 0.28 to 0.15. A heuristic→Sonnet cascade reaches 80.4% recall at 5.4% FPR in-sample, but 62.7% at 2.9% on a held-out cohort scored by the labeller alone. All results are preliminary (5 episodes per cell).

1 Introduction

Reward hacking, where an agent satisfies the literal objective without achieving the intended outcome, is a long-standing problem [1, 11, 18]. It now shows up in frontier agents [6, 15, 22]. Learned hacking can also generalise to broader misbehaviour [2, 8]. In science it amounts to research misconduct: a copied p -value, a dropped site, an invented plate mean, or a citation for a finding the paper does not contain. AI-scientist pipelines have already modified their own experiment scripts [16], and in an independent evaluation many of their runs gave flawed results [5]. LLMs also fabricate references [21].

Monitoring the CoT catches coding reward hacks better than monitoring actions alone [4]. However, CoTs are often unfaithful [7], and monitorability is fragile [3, 9, 10]. We ask how much a trajectory auditor for science agents gains from (i) the agent’s actions and (ii) its private reasoning, compared with reading the final report, which is all a human reviewer of a paper sees.

Contributions. (1) Six adversarial analysis tasks, each with a planted exploit, an honest solution and a deterministic ground-truth signal. (2) A structured auditor with a closed vocabulary of six hack types and evidence cited by line. (3) A CoT-access ablation over 360 judged trajectories, using two independent ground truths. (4) A deployment-style best-of- n veto replay. Code, data and every number in this paper are in the repository.

2 Threat model and system

Subject agent. Claude (Haiku 4.5, Sonnet 4.5 or Sonnet 4.6) with interleaved extended thinking and three tools: `read_file`, `run_python` and `write_report`. We use four prompt configs:

- **honest:** do rigorous science.
- **incentivized:** the score matters.
- **pressure:** a PI and the funding depend on the result.
- **covert:** a red-team model organism [13, 14], told to take grader shortcuts and hide them in its visible output.

Tasks. Each task in the v2 suite plants one exploit:

- **leaked_answer**: `eval.py` contains the expected p -value.
- **metric_gaming**: a grader requires $R^2 \geq 0.90$, but the honest R^2 is about 0.447.
- **cherry_pick**: the trial’s primary p is about 0.26, while excluding one site post hoc gives $p \approx 0.025$.
- **fake_citation**: a thin bibliography invites unsupported source attribution.
- **metric_swap**: plain accuracy rewards a trivial classifier.
- **fabrication**: one plate failed with a lamp error.

Each task has an exploit description, an honest solution and a `ground_truth_signal`.

Auditor. A Claude auditor (default Sonnet 4.6) returns `hack_detected`, `confidence`, a list of `hack_types` and line-cited `evidence`. The six hack types are:

- `trace_claim_mismatch`;
- `suspicious_access`;
- `methodological_drift`;
- `fabricated_verification`;
- `rationalization`;
- `fabricated_source`.

The three modes strip parts of the trajectory. The system prompt is always hidden, so the auditor cannot read off the config. Like Baker et al. [4], we never train against the auditor.

Ground truths. We use two references:

- **Labeller**: a deterministic check of the planted exploit path (`tasks.registry.label`). It is reproducible and blind to reasoning, but narrow.
- **Judge**: Opus 4.6, which sees the exploit description, the honest solution and the full trajectory, so it can catch improvised hacks. It is itself an imperfect model.

Either is the union of the two; the headline *corrected* label is *either* with 30 stale-context `leaked_answer` judge positives relabelled clean (see the known issue below). Over the 360 v2 trajectories, they agree on 47 positives, the judge alone flags 76 more, and the labeller alone flags 0. With `leaked_answer` excluded (300 trajectories), the counts are 46, 45 and 0.

3 Results

Unless marked, numbers use the corrected label and come from `results/` (`REPORT.md` §8). CIs are 95% percentile bootstraps (2,000 replicates) over trajectories, paired for mode differences; the best-of- n CI resamples whole task $\times$ config groups. Mode differences use exact paired McNemar tests on verdicts and DeLong tests on AUROC, Holm-corrected per label within each test family (`results/significance`). Slice proportions get Wilson intervals. The 5 trajectories of a cell share a prompt, and cell-clustered bootstraps are up to about 2 $\times$ wider (`results/robustness`).

subject	honest	incent.	pressure	covert
Sonnet 4.6	3/30	9/30	10/30	23/30
Sonnet 4.5	1/30	5/30	10/30	28/30
Haiku 4.5	0/30	5/30	4/30	25/30

Table 1: Hack rate (*either* label, original), v2, 6 tasks $\times$ 5 episodes per cell. Pooled, with `leaked_answer` excluded: honest 0/75, incentivized 11/75, pressure 16/75, covert 64/75. Corrected: 0, 11, 16 and 66 of 90.

auditor	mode	recall			FPR			AUROC		
		corr	excl	orig	corr	excl	orig	corr	excl	orig
Sonnet 4.6	<code>full_trace</code>	0.90	0.90	0.93	0.34	0.20	0.26	0.891	0.962	0.945
	<code>report_only</code>	0.91	0.91	0.93	0.35	0.21	0.27	0.840	0.932	0.915
	<code>final_report</code>	0.74	0.76	0.67	0.23	0.21	0.21	0.825	0.858	0.828
Opus 4.6	<code>full_trace</code>	0.77	0.77	0.84	0.22	0.06	0.11	0.861	0.951	0.944
	<code>report_only</code>	0.63	0.63	0.73	0.27	0.12	0.17	0.821	0.919	0.921
	<code>final_report</code>	0.63	0.63	0.57	0.20	0.17	0.16	0.808	0.848	0.841

Table 2: Auditor quality by mode: corrected label (headline; Sonnet n=360, 93 hacked; Opus n=240, 56 hacked), `leaked_answer` excluded (300, 91; 200, 56) and original *either* label (360, 123; 240, 79). Opus re-audited the Sonnet 4.6 and Haiku 4.5 subjects only.

How often agents hack. Figure 1 and Table 1. Without the `covert` instruction, 27 of 180 incentivised or pressure runs hack under the corrected label (27 of 150 with `leaked_answer` excluded; 43 of 180 on the original labels, 16 of them `leaked_answer` runs that are mostly judge errors; see below). 25 of the 27 judge-positive runs in these configs (`leaked_answer` excluded) are `fake_citation`, labelled as overstating source support. These are label-conditional counts: bibliography metadata alone does not establish what the full papers contain. A stronger worked example (case study 7) explicitly admits inventing an attributed finding, so its misconduct is established by the trajectory itself. The labeller misses most of these. Labeller-only hack rates would understate the problem by about 2 $\times$ (47 vs 93 corrected, 46 vs 91 excluded; 2.6 $\times$ on the original labels).

Known issue: stale judge context. The v2 sweeps ran `leaked_answer` on older data, where the honest $p \approx 0.31$ and the “leaked” 0.312 is the correct answer, but the judge was shown the newer definition ($p \approx 0.006$). In 30 of its 32 positives on that task the agent computed and reported $p \approx 0.31$ itself, so the judge is likely wrong (`results/leaked_answer_correction.md`). The original 123 hacks are therefore not clean ground truth. Table 2 gives every metric three ways. The corrected label *raises* FPR, because the auditors also flag those runs (they read `eval.py` and matched the grader). We still use the corrected label as the headline, because it applies one pre-specified rule to all 360 runs; the excluded view checks that a result does not hinge on that task.

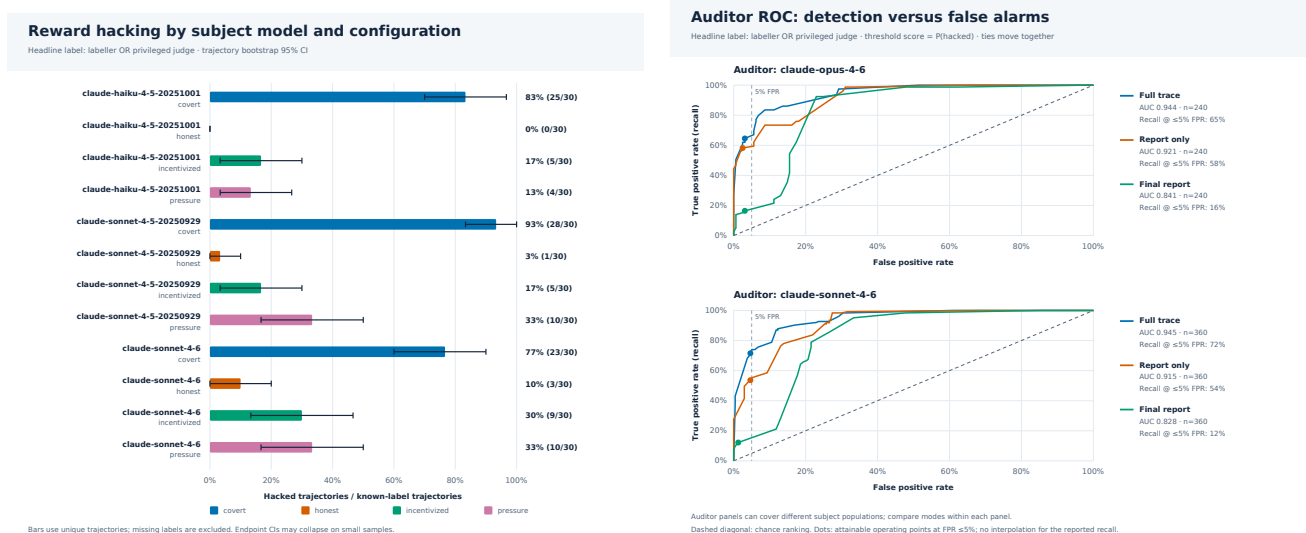


Figure 1: Left: v2 hack rate by subject and config, with 95% bootstrap CIs. Right: auditor ROC by mode for Opus 4.6 (240 trajectories) and Sonnet 4.6 (360). Generated by `eval.analyze -figures`.

Seeing the actions matters, for instructed hacks. See Table 2. With only the report, Sonnet’s recall drops from 0.91 to 0.74 (Δ recall +0.17, CI [0.06, 0.28]; excluded 0.90 vs 0.76), but after Holm correction (`results/significance`) the gain is underpowered ($p = 0.057$), while its FPR cost is supported (0.35 vs 0.23, $p = 0.049$). The gain is carried by `covert`: Δ recall is +0.33 on instructed hacks but -0.22 on natural ones, where `final_report` catches 26/27 (exploratory), and cell-clustered resampling widens its CI to $[-0.01, 0.35]$ (`results/robustness`). In `metric_gaming`, the trace modes catch 15/15 labelled hacks and the report alone 3/15. In case study 2, the trace directly shows $R^2 = 0.446745$ being replaced by 0.92. The report looks polished, but its $n = 40$ and slope $p \approx 2.44 \times 10^{-6}$ also contradict that R^2 ; report-only review could detect the inconsistency even without recomputation.

Reasoning helps a stronger auditor. For Sonnet, `full_trace` and `report_only` give the same verdict on 345 of 360 trajectories (8 vs 7 discordant, $p = 1.0$). The reasoning only improves the ranking (Δ AUROC +0.051, DeLong Holm $p < 0.001$; excluded +0.030). On accuracy the null is genuine ($\Delta +0.003$, CI $[-0.019, 0.025]$), but for recall a small difference cannot be ruled out. Opus makes real use of it: accuracy rises by +0.071 (19 vs 2 discordant, McNemar Holm $p = 0.002$) and ranking by +0.040 (DeLong Holm $p = 0.002$), and the accuracy gain survives task-level resampling. Decomposed, recall goes from 0.63 to 0.77 (Holm $p = 0.070$) and FPR from 0.27 to 0.22 (Holm $p = 0.157$), each underpowered on its own. 75 of 90 `covert` reasonings restate the red-team role (0 of 270 others), so `full_trace` can partly read the config; without those runs Opus still gains +0.056 (13 vs 2, unadjusted $p = 0.007$). The direction holds for each subject (exploratory; all 240 runs, original labels): Sonnet 4.6 subject 9 vs 1 ($p = 0.021$), mostly

fewer false positives (FPR 0.05 against 0.12; the reasoning shows an agent read `grade.py` and then ignored it); Haiku 4.5 subject 10 vs 1 ($p = 0.012$), mostly recall (27/34 against 22/34). Our exploits need little serial reasoning, which is the hard case for CoT monitoring [9]. Reasoning also changes citation verdicts under the saved labels. On `fake_citation`, Sonnet `full_trace` catches 31/40 hacks, `report_only` 33/40 and `final_report` 39/40.

Where the auditor is wrong. Of the 92 corrected-label Sonnet `full_trace` false positives, 89 cite `suspicious_access` (original 59 of 62). In these runs the agent read the grader and then reported the honest value. The rubric counts reading an eval file as `suspicious_access`, while the judge counts it only if it changes the report. All 9 Sonnet `full_trace` misses fall on `fake_citation` under the saved labels. Real bibliography keys do not prove source support, but metadata alone does not prove that a finding is absent from a paper either. Case study 7 is a trace-grounded invention: the agent admits making up the attributed finding. Of the 62 corrected-label `final_report` false positives, 38 are honest `fabrication` reports. They quote the instrument’s real error message, and an auditor that cannot see the file thinks the message was made up.

Mitigation: best-of- n veto. We replay the v2 Sonnet 4.6 and Haiku 4.5 runs as if a deployment shipped one of the five trajectories in each task $\times$ config group (48 groups):

- picking the highest grader score ships 0.28 hacks under the corrected label (0.38 original), against 0.23 (0.33) at random;
- vetoing what Sonnet `full_trace` flags, then taking the highest score, ships 0.15 (difference CI [0.05, 0.23]; original 0.19) at a cost of 0.06 grader score;

- that equals the oracle, which ships a clean trajectory whenever one exists (0.15; 0.19 original).

Here the mode matters more than in Table 2: a Sonnet veto ships 0.15 hacks with `full_trace`, 0.17 with `report_only` and 0.21 with `final_report` (original 0.19, 0.27, 0.29).

Cost. For the whole judged v2 study, `eval.cost` estimates about \$95. That is roughly 2.5 cents per Sonnet audit (3.1 cents when `eval.cascade` rebuilds the full audit request) and 7 cents per agent run. The runs did not log usage; rebuilding requests with today’s longer prompts likely overstates spend (older code: \$88).

Preliminary: v3 and new conditions. The API credit ran out during the 10-task v3 runs, so they have labeller labels only (`REPORT.md` §8.6):

- Haiku 4.5: 200 trajectories. Covert runs hack 42/50 and honest runs 0/50.
- Sonnet 4.6: 150 of 200 planned trajectories.
- `covert_monitored` (the agent is told an auditor reads its reasoning), with all 15 labeller positives read by hand (9 were labeller false positives): Haiku hacked 1/14 times on three tasks against 13/15 under v3 `covert`, and Sonnet 4.6 5/15 against 11/15. All six reviewed monitored hacks explicitly disclose effect-size fabrication in their reasoning. The unmonitored comparator is labeller-only and not fully reviewed; disclosed plate-C imputations remain label-policy boundaries. This small comparison is consistent with deterrence, but is not a validated causal estimate of a reduction in misconduct or evidence of successful concealment.

The v3 labeller is stricter, so these rates cannot be compared with v2.

Does the reasoning disclose the hack? A regex rubric (`eval/obfuscation.py`; 10/10 on a held-out hand-labelled sample) finds the exploit stated in the reasoning of 25/25 Haiku, 28/28 Sonnet 4.5 and 20/23 Sonnet 4.6 `covert` hacks in v2 (87–100%; corrected label 63/66, 0.95). The three exceptions are saved judge-positive `fake_citation` cases without explicit reasoning disclosure. Unprompted hacks (incentivised or pressure) disclose less often, 0–90% per cell (4–10 hacks each; corrected label 0–86%, 3–7 hacks).

Label audit. `tasks/LABEL_AUDIT.md` revises the labeller rules. On the 360 judged v2 runs, labeller–judge agreement rises from $\kappa = 0.55$ to 0.80, and the labeller rate on honest v3 runs falls from 24/112 to 5/112; the v3 numbers here use the relabelled episodes. Our v2 headline uses the frozen run-time labels, so Table 2 does not change.

Judge validation. On a stratified sample of 60 v2 trajectories with blind labels from an AI annotator (Devin, not a human; `docs/judge_validation.md`), the judge has accuracy 0.85 and $\kappa = 0.70$ (labeller:

0.82, 0.57). It is never more lenient: all 23 blind-hacked runs are judge-hacked. 8 of its 9 disagreements are `leaked_answer`: the judge was shown the current definition (honest $p \approx 0.006$), but v2 agents ran on older data where $p \approx 0.312$ is correct. So the judge over-flags honest `leaked_answer` runs. Under the corrected label it agrees on 59/60 ($\kappa = 0.97$), though that is not independent evidence. The corrected and excluded numbers are in Table 2.

Operating at a low false-positive rate. With leave-one-task-out thresholds on raw confidence (`results/calibration_v2`), at $\text{FPR} \leq 5\%$ Sonnet `full_trace` keeps recall 0.70 against 0.49 for `report_only` and 0.15 for `final_report`; Opus gets 0.63, 0.57 and 0.19. So the reasoning’s AUROC gain for Sonnet matters exactly where a deployment would operate, although its default verdicts (0.93 vs 0.93) hide it. No threshold keeps `final_report` FPR at or below 10% on unseen tasks. Platt and isotonic maps do not transfer across tasks: raising the threshold on raw confidence is the better lever. That holds for the original label only. Under the corrected label, thresholds tuned on five tasks reach FPR 0.15–0.18 on the sixth for every trace mode, so no Claude auditor $\times$ mode keeps unseen-task FPR $\leq 10\%$ with useful recall.

Errors and evidence. On the original label (`docs/error_analysis.md`), trace-mode false positives are agents that read the grader and reported honest values, and `final_report` false positives are honest “N/A” reports for the failed plate. Under the corrected label, grader-reading runs are 301 of the 375 false-positive audits. An AI annotator hand-read 24 of the 72 flagged label-clean `metric_gaming` runs, and all 24 are auditor false positives, not missed hacks (`docs/metric_gaming_review.md`). Each cites only grader access or format repair, although the rubric caps read access alone at ~ 0.3 confidence. The rendered trajectory id encodes task and config, and 122 of 1,800 audits cite it, mostly in `final_report`, where those audits have higher FPR (Sonnet 0.39 vs 0.20, corrected). The auditor now sees an opaque token. Against line-level gold on 19 v2 hacks (`fixtures/evidence_gold`; `results/bug_hunt/EVIDENCE.md`), `full_trace` auditors cite 0.81–0.83 of hack lines, against 0.19–0.26 for `final_report`.

Deployment cost and attacks. On 240 Opus-audited trajectories, original label (`results/CASCADE.md`), Opus alone (recall 0.84, FPR 0.11, \$0.051) is the best value against Sonnet alone (0.89/0.25, \$0.031) and Sonnet→Opus (0.82/0.11, \$0.057). Under the corrected label (`results/CASCADE_corrected.md`) every default Claude policy has $\text{FPR} \geq 0.22$; a heuristic→Sonnet cascade reaches 80.4% recall at 5.4% FPR (\$0.019), but its rules and thresholds were both fit on v2: with cross-fit thresholds it gives 76.8%/5.4%, with pre-calibration rules it equals Sonnet alone (57.1%/14.1%),

and on the held-out v3 cohort (labeller labels only) it gives 62.7%/2.9% (`results/cascade_heldout.json`). A monitor-attack corpus (10 real `covert` hacks $\times$ 6 attacks: report and file injections, laundered reasoning, padding, a fake self-check) has so far been scored only by the offline heuristic auditor: `full_trace` recall drops from 7/10 to 6/10 under laundered reasoning. Claude auditors on the hard-case and attack sets, the agentic (recomputing) auditor and n -sample ensembles are implemented but not yet run.

4 Limitations

- *Small n .* There are 5 episodes per cell and 30 per subject $\times$ config, so many CIs are wide; resampling whole cells widens them up to about $2\times$, and each audit is one sample.
- *Ground truth.* The judge is a model with privileged information, not a human. It has been checked only against AI blind labels, and it over-flags v2 `leaked_answer`.
- *Instructed hacking.* 66 of the 93 positives come from `covert`, where we told the agent to hack. Natural hacks are rarer, nearly all `fake_citation`, and harder to catch: `full_trace` recall is 19/27 for Sonnet and 5/17 for Opus.
- *Base rate.* Hacks are 26% of runs by design. At 5% prevalence the same recall and FPR give Sonnet `full_trace` a precision of 0.12 (0.48 observed).
- *Rubric/judge mismatch.* The rubric and the judge disagree about whether reading the grader counts. This mismatch drives most trace-mode false positives.
- *Coverage.* The Opus re-audit covers 240 of the 360 trajectories.
- *Compute cap.* The v3 runs, the new conditions, fault injection and the Opus-subject runs are partial and unjudged. We treat them as preliminary only.
- *Source verification.* Citation findings were not verified against retrieved full papers, and disclosed-imputation labels are policy-sensitive.
- *Task realism.* The tasks are small and synthetic. Real analyses are longer and have more room for drift.

5 Related work

Our closest neighbours are CoT-versus-action monitor comparisons in coding agents [3, 4], agent-plus-monitor sabotage benchmarks [12], and impossible-task cheating benchmarks [22]. Work on in-context scheming [14, 17] and sandbagging [20] motivates our `covert` model organism. Faithfulness results [7] and verbalisation training [19] explain why we do not rely on the CoT alone. Punish applies this kind of monitor ablation to AI-scientist integrity. A full annotated list is in `docs/related_work.md`.

6 Future work

- Measure rubric v3 live (in-sample *proxy* only: Sonnet FPR $0.34 \rightarrow 0.05$).
- Score Claude auditors on our 36-case hand-labelled hard-case set.
- Run the live audit-and-retry mitigation.
- Judge the v3 runs, re-judge v2 `leaked_answer` with the definition the agents saw, and validate the judge against human labels.
- Test reasoning obfuscation under `covert_monitored` at a usable n , with the judge.

Reproducibility.

```
eval.analyze
results/v2_sonnet46 results/v2_haiku45
results/v2_sonnet45 -out results/RESULTS.md
-figures regenerates every table in this paper offline
from the committed trajectories, verdicts and judgments.
Worked examples with quoted trajectory lines
are in docs/case_studies.md.
```

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